Name: Mewada Maitrri Manishkumar

Topic: Network Configuration and Firewall Management in RHEL 9

1. Use the ifconfig or ip command to display the current IP address and network interfaces on your Linux system, then take a screenshot of the output. Display all network interfaces

⇨ Display all network interfaces
ip addr show
or
ip a

```
[root@localhost ~]# ip addr sho
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:e0:da:2f brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.116.128/24 brd 192.168.116.255 scope global dynamic noprefixroute ens160
       valid_lft 1149sec preferred_lft 1149sec
    inet6 fe80::20c:29ff:fee0:da2f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

To display routing information:

ip route

To show interface :

ip link

2. Configure a static IP address for your system using nmcli so that your device always gets the same IP (for example, 192.168.1.50/24 with gateway 192.168.1.1 and DNS 8.8.8.8).

Hint: Use 'nmcli con mod' to set ipv4.addresses, ipv4.gateway, and ipv4.dns, then bring the connection down and up to apply changes.

⇨ Step 1: Find the active connection

nmcli connection show

Example:

```
NAME     UUID                              TYPE      DEVICE
System   xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx  ethernet  ens160
```

Assume the connection name is System.

Step 2: Configure the static IP

```
sudo nmcli connection modify "System" \
ipv4.method manual \
ipv4.addresses 192.168.1.50/24 \
ipv4.gateway 192.168.1.1 \
ipv4.dns 8.8.8.8
```

Step 3: Restart the connection

```
sudo nmcli connection down "System"
sudo nmcli connection up "System"
```

Step 4: Verify

ip addr show

or

nmcli device show

You should now see:

192.168.1.50/24

3. Explain in your own words the difference between DNS and DHCP, and give a real-world example of how each is used in an app like Zomato or Flipkart.

⇨ DNS (Domain Name System) converts easy-to-remember domain names (such as www.flipkart.com) into IP addresses that computers use to communicate.

DHCP (Dynamic Host Configuration Protocol) automatically assigns network settings such as an IP address, subnet mask, gateway, and DNS server to devices when they connect to a network.

Real-world examples

Zomato: When you open the Zomato app, DNS translates www.zomato.com into the server's IP address so the app can connect.

Flipkart: When your laptop connects to Wi-Fi, DHCP automatically assigns an IP address so you can access the internet and browse Flipkart.

4. Enable the firewall on your system using ufw, then allow incoming connections only on port 80 (HTTP) and block all others. Show the status output after applying these rules.

⇨ Step 1: Start the firewall
sudo systemctl enable --now firewalld
Verify:
sudo systemctl status firewalld
Step 2: Set the default zone
(Optional)
sudo firewall-cmd --get-default-zone
Step 3: Allow HTTP
sudo firewall-cmd --permanent --add-service=http
Step 4: Remove other unnecessary services (if required)
For example, if SSH should not be allowed:
sudo firewall-cmd --permanent --remove-service=ssh
Step 5: Reload the firewall
sudo firewall-cmd --reload
Step 6: Check the configuration
sudo firewall-cmd --list-all

5. Use firewalld or iptables to temporarily block all outgoing traffic to a specific website (for example, www.spotify.com) and then verify that you cannot ping the site from your terminal.

Hint: You may need to use the site's IP address instead of the domain name.

⇨ Step 1: Find the IP address
nslookup www.spotify.com
or
host www.spotify.com
Example:

35.186.224.25
(The IP may differ.)
Step 2: Create a rich rule to block that IP

```
sudo firewall-cmd --add-rich-rule='rule family="ipv4" destination
address="35.186.224.25" reject'
```

This is temporary and lasts until the next reboot or firewall reload.
Step 3: Verify the rule

```
sudo firewall-cmd --list-rich-rules
```

Step 4: Test

```
ping 35.186.224.25
```

Step 5: Remove the temporary rule

```
sudo firewall-cmd --remove-rich-rule='rule family="ipv4" destination
address="35.186.224.25" reject'
```